

LIKHITH NAGARALU GURUMURTHY

Boston, MA — +1 (857) 351-9401 — nagaralugurumurthy.l@northeastern.edu
linkedin.com/in/likhith-ng — github.com/LikhithNG

Education

Northeastern University

Boston, MA

Master of Science in Information Systems

Sept 2024 – May 2026

- **Courses:** Data Management & Database Design, Cloud Computing, Application Development, Data Science Engineering Tools.

Visvesvaraya Technological University

Karnataka, India

Bachelor of Engineering in Information Science and Engineering

Aug 2019 – June 2023

- **Courses:** Data Structures, Big Data Analytics, Artificial Intelligence & Machine Learning, Computer Networks, Cloud Computing, Software Engineering.

Technical Skills

Languages: Python (Pandas, NumPy, SciPy), SQL, Java, R, HTML/CSS, JavaScript

Frameworks & Tools: PyTorch, Flask, FastAPI, PyTest, Docker, Jenkins, Git, Tableau, Power BI

Databases: MySQL, MongoDB, Oracle, Snowflake, PostgreSQL

Cloud / QA: AWS (Lambda, EC2), CI/CD, API Testing (Postman), ETL Pipelines, Data Validation

Other: Agile/Scrum, Process Optimization, Statistical Modeling, Technical Documentation

Experience

Data Engineering Intern

Bengaluru, India

Compsoft Technologies (CST)

Aug 2022 – Jan 2023

- Engineered end-to-end **ETL pipelines** in Python and SQL to consolidate structured/unstructured data, reducing data latency by 35%.
- Integrated REST API validation using **PyTest** and **Postman**, expanding test coverage by 40% and improving pipeline reliability.
- Optimized database schema, query indexing, and joins, reducing discrepancies by 25% and improving throughput.
- Implemented **data validation layers**, automated error logging, and CI/CD-ready workflows for data governance.

Quality Analytics Intern

Bengaluru, India

Pro-Test Private Limited

Sept 2023 – Feb 2024

- Developed Python-based automation to monitor **CAPA compliance metrics** and defect reporting, enhancing traceability by 25%.
- Connected **Power BI dashboards** to Salesforce and SharePoint data for live visualization of audit trends and NC performance.
- Performed SQL-driven data audits and statistical risk modeling to uncover systemic quality deviations.
- Led validation and testing of QMS automation pipelines under ISO documentation standards ensuring audit compliance.

Academic Projects

Self-Supervised GNN for Seizure Analysis

PyTorch, PyG, MNE-Python, DCRNN, GAT

- Designed a **graph neural network** pipeline integrating DCRNN and contrastive self-supervised learning for EEG-based seizure classification.
- Implemented dynamic correlation graphs and **multi-head attention** for interpretable spatiotemporal feature extraction.
- Automated EEG preprocessing and normalization using MNE-Python, reducing model training time by 20%.
- Achieved 90%+ F1 and AUROC through optimized **PyTorch Geometric** training and hyperparameter tuning.

Smart Hydroponic Farm Management System

Oracle PL/SQL, SQL Developer

- Architected a relational **3NF schema** with triggers, stored procedures, and constraints to automate greenhouse management workflows.
- Implemented **real-time data ingestion** for crop registration, irrigation monitoring, and inventory tracking.
- Developed SQL-based analytical views for yield forecasting and sensor anomaly detection.
- Enhanced operational efficiency by 30% through database-driven automation and workflow optimization.

Deliver Easy – Enterprise Logistics Platform

Java, MySQL, MVC Architecture

- Developed a full-stack **logistics orchestration system** supporting real-time route optimization and automated dispatch.
- Implemented **multi-role authentication** (admin, driver, customer) and asynchronous job handling using Java concurrency APIs.
- Designed modular MVC components to streamline business logic, reducing query load by 25%.
- Integrated analytical dashboards to track delivery KPIs, efficiency, and route performance metrics.

Leukemia Cancer Detection Using Deep Learning

PyTorch, YOLOv5/v4, Faster R-CNN, Flask

- Built a multi-class **leukemia subtype detection model** using YOLOv5/v4 and Faster R-CNN architectures for microscopic imagery.
- Created a modular data pipeline for augmentation, preprocessing, and hyperparameter tuning.
- Applied transfer learning to improve model convergence speed and enhance detection precision.
- Deployed a Flask-based inference API enabling real-time diagnostic predictions.

Options Risk Monitoring System

Python, Pandas, NumPy, Snowflake, yfinance API

- Engineered a scalable **options analytics engine** automating Greek computation via the Black-Scholes model.
- Created event-driven alert modules for volatility spikes and earnings proximity using Python agents.
- Implemented fault-tolerant ETL workflows with dual persistence in **Snowflake** and CSV.
- Delivered automated end-of-day summaries to improve trading insight and risk transparency.

Research Funding Analytics Dashboard

Tableau, Power BI, Excel

- Designed an interactive **data visualization dashboard** replicating Salesforce-style analytics for research funding metrics.
- Automated data cleaning and normalization workflows across multiple departmental sources.
- Visualized key performance indicators using Tableau and Power BI for leadership decision-making.
- Enabled trend analysis and forecasting to identify funding gaps and partnership performance.

Certifications & Publications

Salesforce AI Associate Certification — Salesforce, Jan 2025.

M001: MongoDB Basics — MongoDB University, Dec 2021.

NoSQL Database & Big Data (Hadoop) — Great Learning Academy, 2022–2023.

Blockchain Technology Workshop — CyberPeace Foundation, 2021.

Publication: “Leukemia Cancer Detection Using Deep Learning,” *IJRMETS*, Vol. 05, Issue 05, May 2023.